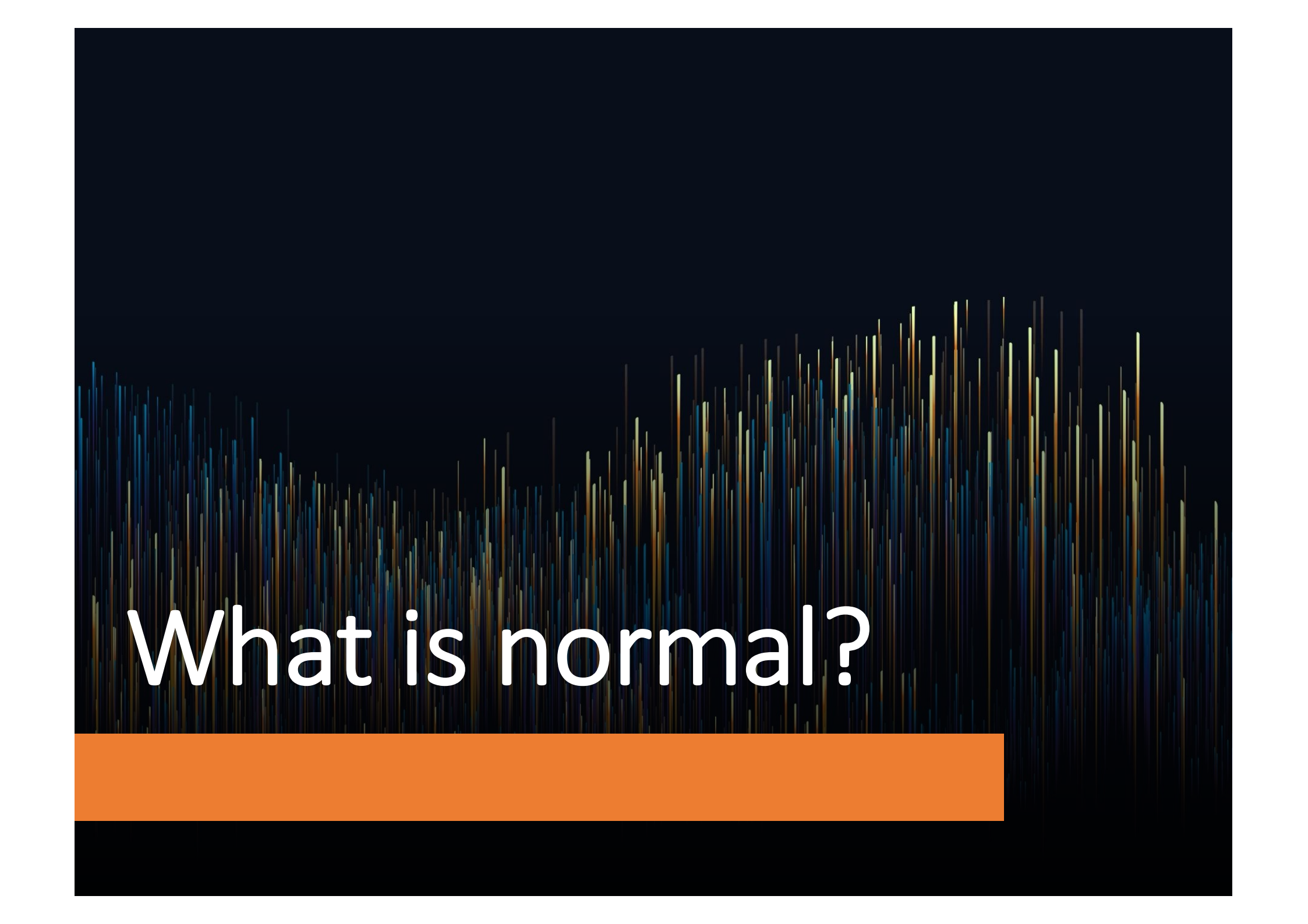




*Module #4*  
Detecting abnormal markets:  
Early Warning Systems & C

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The background is a dark navy blue. It is filled with a dense forest of thin, vertical lines of varying heights and colors, including shades of blue, teal, and gold. These lines create a textured, almost pixelated effect. At the bottom of the image, there is a solid, horizontal orange bar that spans across the width of the frame.

What is normal?

# Anomaly Detection: rationale

Many Fintech activities involve the financial market

This includes, for example:

- robo-advisory
- robo-for-advisory
- trading platforms

And speaking of the financial market means speaking of risks

Market  
crashes do  
happen

### Largest Real Declines in U.S. Stock Market History

| Pain Rank | Pain Index (%) | Peak     | Trough   | Recovery | Decline Rank | Decline (%) | Event(s)                                             |
|-----------|----------------|----------|----------|----------|--------------|-------------|------------------------------------------------------|
| 1         | 100.00         | Aug 1929 | May 1932 | Nov 1936 | 1            | 79.00       | 1929 Crash & Great Depression                        |
| 2         | 89.34          | Jun 1911 | Dec 1920 | Dec 1924 | 4            | 50.96       | WWI & Influenza                                      |
| 3         | 85.51          | Aug 2000 | Feb 2009 | May 2013 | 2            | 54.00       | Lost Decade (Dot-Com Bust & Global Financial Crisis) |
| 4         | 80.41          | Dec 1972 | Sep 1974 | Jun 1983 | 3            | 51.87       | Inflation, Vietnam, & Watergate                      |
| 5         | 59.57          | Feb 1937 | Mar 1938 | Feb 1945 | 5            | 49.93       | Great Depression & WWII                              |
| 6         | 29.06          | May 1946 | Feb 1948 | Oct 1950 | 6            | 37.18       | Postwar Bear Market                                  |
| 7         | 14.22          | Nov 1968 | Jun 1970 | Nov 1972 | 7            | 35.54       | Inflationary Bear Market                             |
| 8         | 8.23           | Jan 1906 | Oct 1907 | Aug 1908 | 8            | 34.22       | Panic of 1907                                        |
| 9         | 8.18           | Apr 1899 | Jun 1900 | Mar 1901 | 9            | 30.41       | Cornering of Northern Pacific Stock                  |
| 10        | 7.73           | Aug 1987 | Nov 1987 | Jul 1989 | 10           | 30.21       | Black Monday                                         |
| 11        | 6.25           | Nov 1886 | Mar 1888 | May 1889 | 13           | 22.04       | Depression & Railroad Strikes                        |
| 12        | 5.00           | Apr 1903 | Sep 1903 | Nov 1904 | 14           | 21.67       | Rich Man's Panic                                     |
| 13        | 4.80           | May 1890 | Jul 1891 | Feb 1892 | 17           | 20.11       | Baring Brothers Crisis                               |
| 14        | 3.55           | Dec 1961 | Jun 1962 | Apr 1963 | 12           | 22.80       | Height of Cold War & Cuban Missile Crisis            |
| 15        | 3.20           | Aug 1897 | Mar 1898 | Aug 1898 | 15           | 21.13       | Outbreak of Boer War                                 |
| 16        | 3.14           | Oct 1892 | Jul 1893 | Mar 1894 | 11           | 27.32       | Silver Agitation                                     |
| 17        | 3.11           | Sep 1909 | Jul 1910 | Feb 1911 | 16           | 20.55       | Enforcement of Sherman Antitrust Act                 |
| 18        | 1.00           | Dec 2019 | Mar 2020 | Jul 2020 | 18           | 20.00       | COVID-19 Pandemic                                    |

Data as of Feb. 28, 2021. Sources: Kaplan et al. (2009); Ibbotson (2020); Morningstar Direct; Goetzmann, Ibbotson, and Peng (2000); Pierce (1982); [www.econ.yale.edu/~shiller/data.htm](http://www.econ.yale.edu/~shiller/data.htm), Ibbotson Associates SBBI US Large-Cap Stock Inflation Adjusted Total Return Extended Index.

Source: <https://www.morningstar.com/articles/1028407/in-long-history-of-market-crashes-coronavirus-crash-was-the-shortest>

## Simultaneous tail events: everything goes wrong



**Frequency**  
of simultaneous  
market crashes



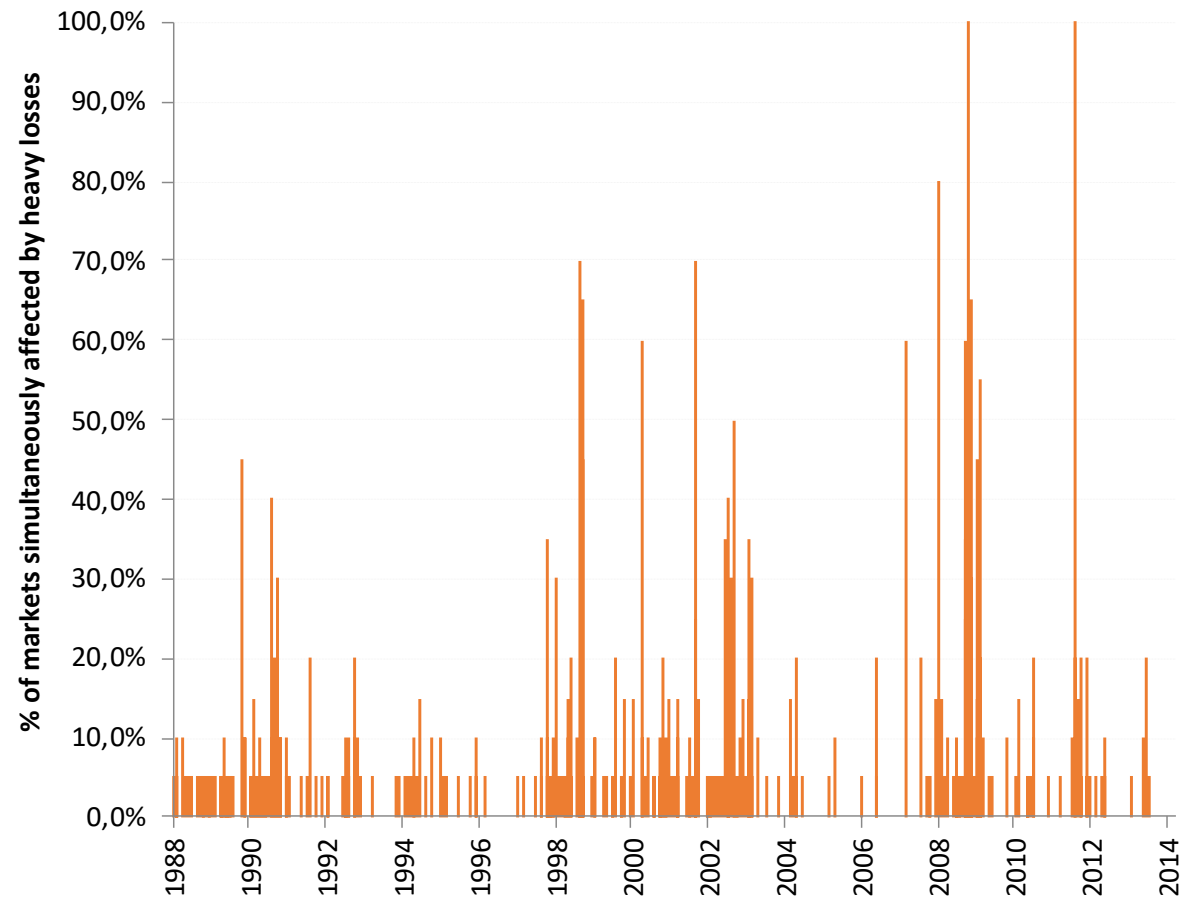
**Cross-asset**  
correlations  
during crises



**Diversification**  
when you  
need it most

The frequency of simultaneous "tail" events is increasing over time

% of markets (all the 20 main world Stock Exchanges) simultaneously affected by large weekly losses (2.5%-tail of the empirical Copula estimated on weekly data since 1/1988)



Source: Zenti, R. (2014) «Volatility, decision models and complexity in financial market», Artificial Intelligence and Cognitive Science, Il Mulino

# The market's worst days had a big impact on investment returns

## The market's worst days have had a large effect on returns

Growth of \$1 in the S&P 500 Index from Dec. 31, 1927, to Dec. 31, 2015

| Days                           | Ending value (\$) | Cumulative return (%) |
|--------------------------------|-------------------|-----------------------|
| Total cumulative return        | 115.40            | 11,440.46             |
| Miss 10 best                   | 38.28             | 3,728.33              |
| Miss 10 worst                  | 362.01            | 36,100.79             |
| Miss 10 best and miss 10 worst | 120.09            | 11,908.91             |
| Cash                           | 19.33             | 1,832.88              |

3X

Sources: Bloomberg L.P., Invesco, Morningstar.

# The problem

- One of the biggest problems of financial market is its annoying tendency to crash
- Market crises correspond to "risk-off" situations, in which risk premia and financial assets exhibit anomalous behavior
- There are big gains in detecting such crashes early on: risk prevention and improved financial performance
- Rather than predicting risk-off situations, it is sufficient to recognize them at their dawn
- It's more nowcasting than forecasting
- The large amount of financial data available invites us to solve the problem using data science



# Solution: Early Warning Systems

- Main goal: detecting crises before most damage has been made; and reducing false alarms
- Data Science provides us with many methods that can be successfully applied (also used in combination)

# Risk-on vs Risk-off

## RISK-ON

### **Investors are confident.**

They bid up risky assets: equities, corporate bonds, EM debt, commodities.

### **Volatility is low. Spreads compress.**

Capital flows toward growth.

## RISK-OFF

### **Investors are scared.**

They sell risky assets and flee to safety: cash, short-term govies, gold, CHF, JPY.

### **Volatility spikes. Correlations surge.**

This is systemic risk.

💡 *Our goal: detect the transition from green to red – as early as possible*



Business  
case:  
Let's look at  
the data

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# Data overview

## Weekly data from Bloomberg

- Key equity indices
- Bond indices (Global, Corporate IG/HY, Inflation-linked, Municipals, Mortgages)
- Short/medium/long term interest rates
- Key exchange rates
- Commodities
- Leading indicators (Economic surprise, Baltic Dry Index)
- VIX (option implied volatility)

A label «abnormal/normal»



# My focus: anomaly detection

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- «Anomaly»: an observation which deviates so much from the other observations as to create suspicion that it was generated by a different mechanism
- Often indicative of something interesting
- Any deviations from the normal behavior that is unusual and significant is of special interest and may require action
- Detecting anomalies is increasingly becoming the core of many business operations, including finance
- Anomaly detection is also used for *data cleaning* — removing outliers from a dataset before training another model – and for *unbalanced classification*

## Same model, two very different business cases

### RISK MANAGEMENT

Reduce drawdown

Protect the portfolio when markets turn toxic

Defensive reallocation

Cash, short-term govies, gold, low-beta assets

Typical client: CIO, risk officer, wealth manager

### QUANT STRATEGY

**Generate alpha**

Aggressive in risk-on, defensive in risk-off

**Dynamic allocation**

Systematic rules, no emotion, replicable, back-testable

Typical client: hedge fund, systematic asset manager

**Which one are you building?** ← This choice shapes everything: data, model, threshold, evaluation metric

# Some basic notions

- **Abnormal** instances = *anomalies* = *outliers*
- **Normal** instances = *inliers*
- **Novelty Detection** = the Anomaly Detection algorithm is trained on a clean dataset - *without outliers* - in order to identify whether a new instance is an outlier or not
- **Outlier Detection** = the Anomaly Detection algorithm is trained on a dataset *with outliers*

# Some applications in Financial Services

Early Warning Systems / Risk  
Modeling

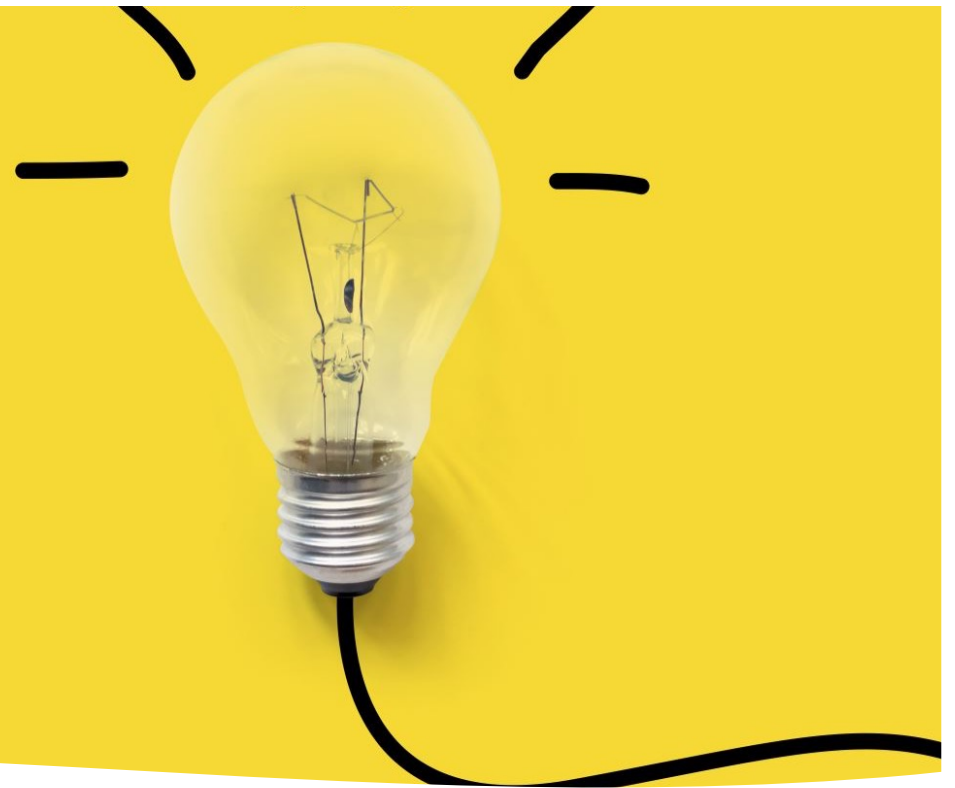
Statistical Arbitrage  
(investments)

Transactional Frauds

Anti-Money Laundering

Customer Behavior Analytics (in  
general)





To give you an  
idea of where  
Anomaly  
Detection can  
go

- In our Recommender System for financial advisors, we are working on looking at all the data relevant to a client, ie:
  - his profile
  - the performance of her portfolios
  - the products she has,
  - etc, also in relation to clusters of similar clients
  - ...
- identifying situations that require immediate attention



Coding session starts |